SECTION 211

PORTLAND CEMENT CONCRETE

211.01 General – This Section describes the requirements for Portland Cement Concrete. Requirements include quality control and quality assurance, constituent materials, mix design proportioning, sampling and testing of the materials and concrete, batching, mixing, transporting, placement, finishing and curing. The Contractor shall supply all necessary materials.

211.02 Applicable Standards – The following standards shall apply unless specified otherwise herein or in the Special Provisions or Drawings. All referenced standards shall be to the current editions at time of Tender.

(a) CSA A23.1 Concrete materials and methods of concrete construction

(b) CSA A23.2 Test methods and standard practices for concrete

(c) CSA A3000 Cementitious Materials Compendium

(d) CSA S269.1 Falsework and formwork

(e) CSA A283 Qualification Code for Concrete Testing Laboratories

211.03 Concrete Mix Designs and Submission Requirements

211.03.01 General – The Contractor shall be responsible for quality control of all components of the concrete operation, including but not limited to, aggregate and component quality, proportioning, test batching, batching, mixing, transporting, placing, consolidating, finishing, curing, and all necessary quality control and verification testing of the components and the fresh and hardened concrete.

211.03.02 Proportioning and Qualification – The Contractor shall be responsible for proportioning and designing all concrete in full compliance with the required concrete mix parameters as listed in the project Special Provisions or Drawings. All mix designs shall use the "Absolute-Volume Method" for mix proportioning.

Sampling and testing of concrete for purposes of qualifying materials and mix designs shall be in accordance with CSA A23.2-24C. Qualification testing requirements are presented in Table 211-A.

211.03.03 Alkali-Aggregate Reactivity (AAR) Preventative Requirements – Total alkali content (Na_2O equivalent) contributed by the Portland cement to the concrete mix shall be in accordance with CSA A23.2-27A. Maximum replacement levels of supplementary cementitious materials shall be in accordance with the Special Provisions. Aggregates with a classification of "extremely reactive" shall not be used.

The Ministry Representative reserves the right to alter this limit or to redefine the concrete mix design parameters based on the review of the mix design submittal.

Unless otherwise specified, the classification of structures shall be St4 in accordance with the CSA A23.2-27A Table 5 except as follows:

(i) Curb, sidewalks and gutters shall be St2; and

(ii) Roadside barriers shall be St3.

Table 211-A: Qualification Requirements

Compressive Strength	All concrete mixes
Slump and/or Slump flow test	All concrete mixes
Air content of plastic concrete	All concrete mixes
Air void system in hardened concrete	All concrete mixes
Resistance to sulphate attack	Additional requirements for concrete subjected to sulphate attack during service shall be in accordance with CSA A23.1, Table 3. The maximum expansion of total cementitious materials when tested using CSA A3004-C8 shall be 0.05% at 6 months or 0.10% at 1 year for S-1 and S-2 exposures. The maximum expansion shall be 0.10% at 6 months for S-3 exposure.
Chloride ion penetrability	All concrete mixes containing any steel reinforcement and exposed to chlorides during service
Linear shrinkage	All cast in place concrete mixes for bridge decks. Not required if the deck concrete mix incorporates ≤ 400 kg/m³ of cementitious materials and a nominal maximum size coarse aggregate of 28 mm (or larger).

211.03.04 Mix Design Submittals – The Contractor shall submit a report outlining the proposed mix design for each classification of concrete to the Ministry Representative for review and acceptance at least 2weeks in advance of when concrete production is scheduled to commence.

Acceptance of the mix design by the Ministry does not constitute acceptance of the concrete. Acceptance of the concrete will be based on the test results and the performance and quality of the concrete and concrete components placed in the Work. No concrete shall be placed prior to receiving Ministry acceptance of the mix design.

Each mix design submittal shall include all of the following (Incomplete submittals by the Contractor may result in rejection of the mix design submittal):

(a) Project Specifics

(i) Name and location of the proposed supplier.

(ii) Distance and expected travel time from batch plant location to project Site.

(iii) Expected method of batching, transporting and placing concrete.

(iv) Specified mix parameter requirements as listed or referenced in the project's Special Provisions or Drawings.

(v) Name and contact information of certified quality control testing laboratory and certified testing personnel. Certification shall be in accordance with the requirements of CSA A23.1.

(b) Materials – All materials shall comply with the requirements as listed in SS 211.04.

(i) Cementitious Materials – Types and source of each material including mill test reports and manufacturer's certificates of compliance.

(ii) Aggregates – Type and source of all individual aggregate products including individual gradations and all other aggregate quality testing as described in Table 211-B for each product and each aggregate source.

(iii) Admixtures – Type and source of all admixtures, including expected dosage rates, point of addition to the mix, compatibility documentation, and individual technical data sheets.

(iv) Water – The source of mixing water. (depending on the source, the Ministry Representative may request additional test documentation).

(v) Miscellaneous – Documentation for all other materials proposed for the mix, showing conformance with applicable Ministry requirements and manufacturer/industry guidelines and standards.

(c) Proportioning

(i) The concrete proportions expressed in terms of quantity of each component.

(ii) Fresh and hardened concrete properties.

(iii) The mass of materials in kilograms (kg) and the absolute volume in cubic metre (m3) contributed by each material in the mixture.

(iv) The quantity of each individual cementitious material added in a powder state shall be expressed as kilograms per cubic metre (kg/m3).

(v) Materials added in slurry shall have their respective solid and water contents expressed as kilograms per cubic metre (kg/m3).

(vi) The quantity of each separately batched size of coarse aggregate and fine aggregate shall be expressed as kilograms per cubic metre (kg/m3) in a saturated-surface-dry (SSD) condition.

(vii) The quantity of water shall be the maximum amount allowed based on the maximum specified water-to-cementitious material ratio (w/cm), corrected to account for any additional water that may be contributed by other materials used in the proposed mix.

(viii) The quantity of water shall be expressed as kilograms (or Litres) per cubic metre (kg/m3 or L/m3). If ice is used for temperature control, it shall be expressed in incremental units (sacks, bags, or pails, etc.) and kilograms per cubic metre (kg/m3) and shall be included in the maximum allowable water quantity.

(ix) Admixtures dispensed as liquids shall be expressed as Litres per cubic metre (L/m3) and where applicable, expected dosage range stated.

(x) The quantity of any pre-measured, pre-packaged additives, such as fibres, shall be expressed in incremental units (sacks, bags, or pails, etc.) and kilograms per cubic metre (kg/m3).

(xi) The absolute volume of each material, air content and the total sum of the absolute volumes of all materials shall be provided in cubic metres (m3).

(xii) Slump, unit weight, air content, strength characteristics, and any other specified characteristic shall be reported for each set of mixture proportions intended for use.

211.03.05 Trial mix test batches – The Contractor shall produce test batches of 4.0 m^3 volumes on all proposed bridge deck and deck overlay mix designs. The Contractor shall produce test batches of 4.0 m^3 volumes on other mix designs where deemed necessary by the Ministry Representative.

Table 211-B: Required Aggregate Testing for Normal Density Coarse and Fine Aggregate (Per Individual Product & Aggregate Source)

Test Method	Test Description	Test Data Validity [5]
CSA A23.2-2A[1] CSA A23.2-5A[1]	Sieve Analysis of Fine and Coarse Aggregate Amount of Material finer than 80 µm in Aggregate	Within 90 days[6]
CSA A23.2-3A	Clay Lumps in Natural Aggregate	Within 3 years
CSA A23.2-3B	Determination of total or water-soluble sulphate ion content of soil	Within 3 years
CSA A23.2-4A	Low Density Granular Material in Aggregate	Within 3 years
CSA A23.2-4B([10])	Sampling and determination of water-soluble chloride ion content in hardened grout or concrete	Within 3 years
CSA A23.2-6A, 12A	Relative Density and Absorption of Fine and Coarse Aggregate	Within 1 year
CSA A23.2-7A	Test for Organic Impurities in Fine Aggregates for Concrete	Within 2 years
CSA A23.2-8A[3]	Measuring Mortar-Strength Properties of Fine Aggregate	Within 2 years
CSA A23.2-9A[2]	Soundness of Fine and Coarse Aggregate by Use of Magnesium Sulphate	Within 3 years
CSA A23.2-13A	Flat and Elongated Particles in Coarse Aggregate - Procedure A (Length to Width Ratio 4:1)	Within 3 years
CSA A23.2-14A[9]	Potential Expansivity of Aggregates (Procedure for Length Change Due to Alkali-Aggregate Reaction in Concrete Prisms at 38°C)	Within 3 years
CSA A23.2-15A[4]	Petrographic examination of aggregates	Within 3 years
CSA A23.2-16A[7]	Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine	Within 3 years
CSA A23.2-23A[2]	Test Method for the Resistance of Fine Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus	Within 3 years
CSA A23.2-24A[8]	Test Method for the Resistance of Unconfined Coarse Aggregate to Freezing and Thawing	Within 3 years
CSA A23.2-25A[11]	Test Method for Detection of Alkali-Silica Reactive Aggregates by Accelerated Expansion of Mortar Bars	Within 3 years
CSA A23.2-29A[2]	Test Method for the Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus	Within 3 years

1. *Each aggregate product that may be used as a component of the proposed mix must have a washed sieved analysis performed showing the cumulative percent passing for each of the following sieve sizes; 56, 40, 28, 20, 14, 10, 5, 2.5, 1.25, 0.630, 0.315, 0.160 and 0.080 mm. Sieve analysis reports are to include all relevant sampling information (date/sampled by/location/field sample size) and include the actual test specimen sample mass.*
2. *Subject to approval by the Ministry Representative, Test Method A23.2-9A can be performed as an alternative to performing Test Method A23.2-23A and A23.2-29A. Refer to CSA A23.1, Section 4.2, Table 12 for further information.*
3. *Test Method A23.2-8A is required only if the aggregate fails to meet the requirements of Test Method A23.2-7A as described in CSA A23.1 Section 4.2.3.3.3.2*
4. *Petrographic Examination of Aggregates: The Contractor shall submit the results of the Petrographic Analysis, certified by a qualified professional registered with the Association of Professional Engineers and Geoscientists of the Province of B.C. and indicating the aggregate's suitability for its intended use. Test shall be performed on coarse and fine aggregates. The petrographic analysis shall not be used to forfeit the requirement of performing other necessary aggregate quality testing.*
5. *Test data validity requirements are based on time periods prior to the expected initial date of concrete production. Test validity refers to time from completion of the tests, subject to verification that the sample tested is representative of production proposed for qualification or acceptance. The Ministry Representative, based on submitted test data and/or visual observations, may waive or alter the above listed time periods. For multiyear projects, test data validity shall be maintained relative to the date of concrete placement.*
6. *Quality control during aggregate production shall include the following test methods at the specified test frequencies: CSA A23.2-2A & A23.2-5A (1 per 300 tonnes). Testing shall be completed on a blend of aggregate reflecting the mix design.*
7. *Subject to approval by the Ministry Representative, Test Method CSA A23.2-16A can be waived if CSA A23.2-29A test results meet the specified criteria.*
8. *This test shall be required should any of Test Methods 4A, 9A, 16A or 29A fail to satisfy the allowable limits. Test shall be performed on both coarse and fine aggregates or a blend of aggregates representative of the proposed mix.*
9. *Testing is waived if CSA A23.2-25A test results classify the aggregates as non-reactive.*
10. *ASTM C1218 may be used in place of A23.2-4B*
11. *Test shall be performed on both coarse and fine aggregates or a blend of aggregates representative of the proposed mix*

Point of discharge sampling and testing shall be carried out by the Contractor to verify the pertinent parameters of the proposed mix design. If placement is by pump, concrete shall be sampled and fully tested at both the truck chute discharge and at the pump discharge locations. Sampling and testing shall be in accordance with SS 211.09, Table 211-C, and Table 211-D.

Test parameters include all required properties of the plastic and hardened concrete, workability and finishability of the mix, strength gain characteristics, including 7 and 28-day compressive strengths, and any other additional testing of the hardened concrete that may be required by the Ministry Representative. Batching, haul times, conveyance procedures, placement and finishing used for the test batch must emulate the procedures expected during the actual concrete placement.

When set retarders or HSAs are used, extra testing shall be carried out to verify the extended slump life and set time. Set time testing shall be carried out in accordance with ASTM C403.

The Contractor shall be fully responsible for all costs associated with the test batches. Test batch trials that do not meet requirements shall be repeated.

211.03.06 Adjustments to Mix Designs – If, during the progress of the work, the mix design is found to be non-compliant or demonstrates potential non-compliance(s) with specifications, the Contractor shall make the necessary adjustments. Adjustments shall be accepted by the Ministry Representative prior to subsequent concrete placement. Notwithstanding the Ministry Representative's review and acceptance of the proposed mix design, it remains the Contractor's responsibility to ensure that the concrete meets all Contract requirements.

MATERIALS

211.04 Materials

211.04.01 General – All material properties used in concrete production shall conform to CSA A23.1, unless otherwise specified in these specifications. Laboratory testing shall be carried out by a testing laboratory certified in accordance with the requirements of CSA A283. Field test procedures shall be undertaken by experienced personnel certified in accordance with CSA A23.1.

211.04.02 Cementitious Material – The Contractor shall obtain, from the supplier, a mill test report and a manufacturer's certificate of compliance representing each type of cementitious material for each month period representing the delivered shipment.

(a) **Hydraulic Cement** – Unless otherwise specified, the Contractor shall use Type GU (General Use) or Type GUL (General Use Limestone) Cement in all concrete. Cement shall conform to the requirements of CSA A3000. GU and GUL cements shall have a C3A content greater than 4.0% (by standard Bogue calculation) or 5% by mill certificate.

The total alkali content [Na_2O equivalent, (Na_2O + (0.658 x K_2O))] of the cement shall not exceed 0.60% by mass.

(b) **Silica Fume** – Condensed silica fume shall conform to the requirements of CSA A3000 for Type SF silica fume.

(c) **Fly Ash** – All fly ash shall conform to the requirements of CSA A3000 for Type F fly ash.

(d) **Slag –** All slag shall conform to the requirements of CSA A3000 for Type S ground granulated blast furnace slag.

(e) **Blended Hydraulic Cement –** Blended Hydraulic Cement shall not be used without the acceptance of the Ministry Representative.

211.04.03 Water – Water to be used for mixing concrete or mortar and for curing shall conform to the requirements of CSA A23.1 and shall be free from contamination by oil, acid, alkali, organic matter, sediment or other deleterious substances. The Contractor shall not use water from shallow, stagnant or marshy sources.

211.04.04 Admixtures – All admixtures shall conform to CSA A23.1.

Admixtures shall be incorporated into the mix in accordance with the mix design that has been reviewed and accepted by the Ministry Representative.

All concrete shall contain Air-Entraining Admixtures. Air-Entraining Admixtures shall conform to ASTM C260.

Chemical admixtures shall conform to ASTM C494.

ASTM C494 Type F High Range Water Reducer (superplasticizer) shall be used when concrete contains silica fume.

All accepted admixtures shall be compatible with all other admixtures and constituents used in the mix.

Calcium chloride or any admixture containing chloride ions shall not be used. If requested, a written statement from the manufacturer stating that the admixture contains no intentionally added calcium chloride shall be provided to the Ministry Representative.

211.04.05 Aggregates – All aggregates are to be natural, processed or manufactured granular material composed of hard, sound and durable particles, free of adherent coatings, shale, clay, organic materials, and other soft or disintegrated pieces.

(a) Coarse Aggregate – All coarse aggregate materials shall conform to the following requirements:

(i) Sampling shall be according to CSA Test Method A23.2-1A.

(ii) Gradation shall be determined in accordance with CSA Test Methods A23.2-2A and A23.2-5A. (All gradations shall be based on a washed sieve analysis, see Table 211-B Footnote 1).

(iii) Gradation of the coarse aggregate shall be within the limits shown in Table 211-E for the particular size range or nominal maximum size of coarse aggregate specified.

(iv) The gradation of coarse aggregates used shall be such that the percentage passing any one sieve does not vary more than 5% from the initially approved coarse aggregates used in the mix design.

(v) The maximum Petrographic Number (PN) of coarse aggregate shall be determined in accordance with CSA A23.2-15A and shall not exceed 125.

(vi) The amount of deleterious substances in coarse aggregate shall not exceed the limits prescribed in Table 211-F.

(vii) Other specified physical properties measuring aggregate quality referenced in Table 211-B shall not exceed the limits prescribed in Table 211-F.

(b) Fine Aggregate – Fine aggregate shall conform to the following requirements:

(i) Sampling shall be according to CSA Test Method A23.2-1A.

(ii) Gradation shall be within the limits given in Table 211-G when tested in accordance with CSA Test Methods A23.2-2A and A23.2-5A. (All gradations shall be based on a washed sieve analysis).

(iii) Fineness modulus shall be between 2.3 and 3.1.

(iv) Fineness modulus tolerance shall be ±0.2 (except that the above sentence shall apply) from the fineness modulus of the initially approved fine aggregate used in the design mix.

(v) A maximum 45% shall be retained between any two consecutive sieves.

(vi) The amount of deleterious substances in fine aggregate shall not exceed the limits prescribed in Table 211-F.

(vii) Other specified physical properties measuring aggregate quality referenced in Table 211-B shall not exceed the limits prescribed in Table 211-F.

Table 211-C: Standard Test Methods for Sampling and Testing

Test Methodology	Test Description	Minimum Test Frequency
CSA A23.2-1C	Sampling Plastic Concrete	As stated below
CSA A23.2-3C CSA A23.2-9C	Making and Curing Concrete Compression and Flexural Test Specimens Compressive Strength of Cylindrical Concrete Specimens	One set for every 35 m^3 or portion thereafter for each classification of concrete placed within an individual structural element or component, and placed on any one calendar day from a single supplier. The Ministry Representative may request additional sets.
CSA A23.2-4C CSA A23.2-5C and 19C CSA A23.2-17C	Air Content of Plastic Concrete by the Pressure Method Slump and Slump flow of Concrete Temperature of Fresh Concrete	Every individual load unless approved otherwise by the Ministry Representative Retests shall be performed following any load adjustments.
CSA A23.2-6C	Density and Yield of Plastic Concrete	Density and Yield tests shall be performed with every air test and strength test.

Table 211-D: Additional Test Methods

Test Methodology	Test Description	Minimum Test Frequency and Acceptance Criteria
ASTM C457	Microscopic Determination of Parameters of the Air-Void System in Hardened Concrete	One test per individual trial mix test batch for deck overlay and deck concrete. The test sample shall be made by coring from the hardened concrete placed in the test trial. Acceptance criteria shall be based on air content Category 1 in accordance with CSA A23.1 Sections 4.3.3.2 and 4.3.3.3. The procedure used shall be reported. Further testing, during scheduled concrete pours may be required, as directed by the Ministry Representative.
ASTM C1202 or CSA A23.2-23C	Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration	One test per individual trial mix test batch for deck overlay and deck concrete. One test shall consist of the average result from a minimum of two specimens. Not less than one test for each 100 m^3 (or part thereof) for each concrete mix design for reinforced concrete elements exposed to chlorides. A minimum of one test shall be required per day for concrete of a single mix design. Acceptance criteria shall be as follows: • less than 1,000 coulombs at 91 days for late chloride exposure deck and deck overlay concrete (1). • less than 1,500 coulombs at 28 days for early chloride exposure deck and deck overlay concrete (2). • less than 1,500 coulombs at 91 days for other reinforced concrete elements exposed to chlorides. Further testing, during scheduled concrete pours may be required, as directed by the Ministry Representative. When the test result does not meet the criteria as specified above, the Contractor shall prepare a plan for remedial measures and the plan shall be submitted to the Ministry Representative for review and acceptance.
CSA A23.2-21C	Test method for length change of hardened concrete (Drying shall commence after 7 days of curing and shrinkage determined after 28 days of drying)	One set of test per individual trial mix test batch for deck overlay and deck concrete. Shrinkage shall be less than or equal to 0.04% at 28 days. Where shrinkage testing conducted on field-cast samples is used as a basis of acceptance, concrete shall be considered acceptable if the average value of tests conducted is equal to or less than 0.040% with no individual value greater than 0.057% (for 100-mm cross-section concrete prism tests) and equal to or less than 0.050% with no individual value greater than 0.061% (for 75-mm cross-section concrete prism tests). This testing shall be waived if the total cementitious materials content in the concrete mixture is ≤ 400 kg/m^3 with the use of a Nominal Maximum Size of Coarse Aggregate of 28 mm.

1. *Late chloride exposure applies to concrete that is allowed to air-dry for a period of at least one month after the end of the curing period, before exposure to de-icing chemicals.*

2. *Early chloride exposure applies to concrete that is not allowed to air-dry for a period of at least one month after the end of the curing period, before exposure to de-icing chemicals.*

Table 211-E: Coarse Aggregate Gradations

NMSA[1] mm	Product Size mm	Total Passing Each Sieve (Percentage by Mass)								
		56 mm	40 mm	28 mm	20 mm	14 mm	10 mm	5 mm	2.5 mm	1.25 mm
40	40 – 5 [2]	100	95-100	–	35 - 70	–	10 - 30	0 - 5	–	–
28	28 – 5 [2]	–	100	95 - 100	63 - 83	30 - 65	–	0 - 10	0 - 5	–
20	20 – 5	–	–	100	85 - 100	50 - 90	25 - 60	0 - 10	0 - 5	–
14	14 – 5	–	–	–	100	90 - 100	45 - 75	0 - 15	0 - 5	–
10	10 – 2.5	–	–	–	–	100	85 - 100	10 - 30	0 - 10	0 - 5

(1) NMSA – nominal maximum size of coarse aggregate. Definition: The standard sieve size opening immediately smaller than the smallest through which all of the aggregate must pass.

(2) To prevent segregation, aggregates that make up the above gradings shall be stockpiled and batched in two or more separate sizes as per CSA A23.1

Table 211-F: Limits for Deleterious Substances In and Physical Properties of Aggregates

Test Method	Substance or Property	Maximum Percentage by Mass of Total Sample	
		Coarse Aggregate	Fine Aggregate
CSA A23.2-3A	Clay Lumps	0.30	1.0
CSA A23.2-4A	Low-Density Granular Materials	0.5	0.5
CSA A23.2-5A	Material Finer than 0.08 mm for Concrete Subject to Wear	1.0	3.0
	Material Finer than 0.08 mm for Other Concrete	1.0	5.0
CSA A23.2-9A	Magnesium Sulphate Soundness Loss	12	16
CSA A23.2-16A	Los Angeles Abrasion Loss for Concrete Subject to Wear	50 (35 for all deck concrete)	N/A
CSA A23.2-23A and -29A	Micro-Deval Abrasion Loss	17	20
CSA A23.2-12A	Absorption Percentage	1.75	N/A
CSA A23.2-7A	Organic Impurities in Fine Aggregate	N/A	Standard colour or lighter
CSA A23.2-13A	Flat and Elongated Particles in Coarse Aggregate – Procedure A (Length to Width Ratio 4:1)	20	N/A
CSA A23.2-24A	Test Method for the Resistance of Unconfined Coarse Aggregate to Freezing and Thawing	6	N/A

211.04.06 Steel Fibres – Steel fibres shall meet the requirements of ASTM Standard A820 Type 1 Cold Drawn High Tensile Deformed Steel Fibres. A mill certificate showing compliance with ASTM Standard A820 shall accompany each delivered lot.

211.05 Storage and Handling of Materials - General – The storage and handling of all materials used in the production of concrete shall conform to CSA A23.1, unless otherwise specified herein. All materials shall be stored and handled in a manner that will prevent contamination or deterioration, otherwise they shall be rejected. Access shall be provided to the storage facilities to allow for inspection and sampling. Where applicable, materials shall be identified by the Safety Data Sheets (SDS).

Table 211-G: Fine Aggregate Gradation

Sieve Size (mm)	Total Cumulative Passing Each Sieve (Percentage by Mass)
10	100
5	95 - 100
2.5	80 - 100
1.25	50 - 90
0.630	25 - 65
0.315	10 - 35
0.160	2 - 10

211.05.01 Cementitious Material – All cementitious materials shall be stored in a suitable weather-tight structure capable of protecting the materials from dampness, hydration and contamination, otherwise they shall be rejected. Materials shall be free of lumps at all times during their use in the work.

211.05.02 Tanks – Tanks used for storing and/or hauling, concrete mixing and curing water shall be free from contamination by oil, acid, alkali, organic matter, sediment, rust or other deleterious substances.

211.05.03 Admixtures – All admixtures shall be stored in manufacturers' original labelled containers with clearly legible labels and be kept above freezing at all times and in accordance with the manufacturers' technical data sheets. Admixtures damaged by any cause shall be rejected.

211.05.04 Other Materials – Other materials, such as steel fibres, etc. shall be stored and handled in accordance with the manufacturer's recommendations and instructions.

211.05.05 Aggregates – Each nominal size of aggregate, including coarse and fine shall be stockpiled separately, stored and handled in a manner that will prevent contamination, intermixing and segregation. A minimum of 10 m^3 of each aggregate size, in excess of the requirements of the daily concrete, shall be maintained in identifiable stockpiles at the batch site in order to ensure proper continuity of the work with approved aggregates.

Stockpiled aggregate, which is segregated, contaminated or intermixed with foreign matter of any kind, shall be rejected. Separate batch plant storage bins shall be provided for each size of aggregate to be batched.

Precautions shall be taken when moving, handling and transferring aggregates to prevent contamination, segregation or degradation in accordance with the following recommendations:

(a) Aggregate blanket 150 mm thick should be left on the ground as stockpiles are depleted.

(b) Aggregate transfer over stockpiles or storage bins of other sized aggregates should be avoided.

(c) Storage bins should be filled only to 150 mm from the top unless a suitable extension rim is provided to prevent contamination.

(d) Front-end loader buckets should be filled by raising them vertically up the face of the stockpile, not by thrusting into the stockpile in one place.

(e) Depositing aggregates into the batch plant storage bins should be made directly over the bins' outlet.

(f) Storage bins should be kept as full as possible during batching to minimize segregation as the aggregates are withdrawn from the bins.

(g) Acceptable and uniform aggregate moisture content shall be obtained and maintained by necessary means to include the following:

(i) Washed or wetted aggregates should remain in stockpiles for a minimum of 12 hours to permit a uniform moisture content throughout the stockpile to be reached before aggregate transfer to batch plant storage bins.

(ii) After each day's concrete production during wet weather, fine aggregate bins should be either emptied or covered with a suitable waterproof covering.

CONSTRUCTION

211.06 Batching

211.06.01 Batching Equipment – Batching equipment shall comply with the requirements of CSA A23.1.

(a) General – Batching equipment, such as weighing mechanisms, gates, water lines and dispensing systems, to be maintained in good working order. Batching bins to be completely emptied of all material before the first and succeeding batches are measured.

Batching equipment to have the following capabilities and facilities:

(i) bins or silos for storage of aggregates, silos for storage of cement and supplementary cementitious materials.

(ii) free movement of each type or size of material to discharge openings without contamination.

(iii) design of storage and hauling facilities to prevent:

(A) loss or intermingling of different sizes and types of materials.

(B) contamination by deleterious substances.

(C) harmful segregation and breakage.

(iv) covers to stockpiles of fine aggregates to ensure constant moisture content.

(v) all fulcrums, clevises and similar working parts maintained in clean condition.

(vi) provision made, in remote control plants and/or where batch cycles are timed, for the complete filling and discharge of the measuring unit for each batch.

(b) Batch Charging Mechanism – Batch Charging Mechanism requirements to be as follows:

(i) material flow control within the specified tolerances.

(ii) hopper construction to eliminate material accumulation and to permit the complete discharge of every batch.

(iii) cement weighing hopper:

(A) self-cleaning and properly ventilated to allow air to escape.

(B) accessible for inspection.

(C) dust sealed between the charging mechanism and the hopper to maintain weighing accuracy.

(c) Scales and Dispensing Mechanism – Scales and Dispensing Mechanisms to conform to the following requirements:

(i) beam type or springless dial type.

(ii) other methods of weighing (electric, hydraulic, load cells, etc.) are acceptable provided the specified weighing tolerances and accuracy requirements are met.

(iii) all accurate to ±0.4% of the total capacity of the scale when static load tested.

(iv) zero balance adjustment capability.

(v) unaffected by binding or vibration due to vibrators or other appurtenances.

(vi) working range between 10% and 90% of the scale reading.

(vii) equipped with prominent markers with individual batch weight setting capabilities for dial scales.

(viii) calibration facilities including an adequate number of standard test masses.

(ix) beam type scales with provision for indicating to the operator that the required load in the hopper is being approached; the device shall indicate at least the last 100 kg of the load.

(x) all weighing and indicating devices to be in full view of the operator while charging the hopper, and the operator shall have convenient access to all controls.

(d) Volumetric Devices for Water Measurement – Volumetric devices for water measurement to be:

(i) fitted with such valves and connections as are necessary to divert the water measured for a batch to easily verify the accuracy of measurement.

(ii) arranged so that measurements will be unaffected by variable pressures in the water supply line.

(iii) calibrated to an accuracy of 2% of the batch volume.

(e) Dispensing Systems for Liquid Admixtures – Dispensing systems for liquid admixtures, other than weigh systems, to have the following capabilities:

(i) visual, volumetric measuring or readout units located as close as possible to eye level for easy reading by the operator.

(ii) positioned in such a manner that discharge into the batch is observed by the operator from a normal working position.

(iii) volumetric measuring unit for periodic check of dispenser accuracy where a positive displacement system is used.

(iv) visual measuring unit for an accurate visual check of increments of 30 mL of air entraining agent or of chemical required to treat a maximum of 25 kg of cement.

(f) Certificate or Report of Inspection – A certificate or report of inspection, from a Weights and Measures approved, independent testing authority, and which is not more than one year old for the plant in its present position, shall be on display in the plant at all times. The certificate or report will be accepted as proof of accuracy of the scales or weighing devices. Where there is reasonable doubt concerning the accuracy of the scales or weighing devices the Ministry Representative

may require their recalibration and recertification at the Contractor's expense, during progress of the work. Plant relocation or major alterations shall require such recalibration and recertification.

211.06.02 Batching of Materials – Material may be weigh-batched separately or cumulatively.

Aggregates shall be batched by mass, cement and mineral admixtures in the powder form by mass or bag.

The mass of any ingredient to be batched shall not be less than 10% nor greater than 90% of the scale capacity.

When batched by 40 kg bags, only full and sealed bags of cement shall be used with no fractional bags permitted. Cement batched by mass shall be to an accuracy of ±1% for batch quantities between 30% and 90% of the scale capacity. When the mass of cement being batched is between 10% and 30% of the scale capacity, the mass of cement batched shall not be less than the required mass nor more than 4% in excess. Intermediate bulk cement shall be weighed to an accuracy of ±1%.

When aggregates are measured by mass, batch masses shall be based on the required mass of saturated surface dry aggregate corrected for the moisture conditions of the aggregate at the time of batching.

Field determination of free moisture in the aggregates shall be made by any proven method such as CSA A23.2-11A or by the use of a device such as the "Speedy Moisture" gauge. When doubt exists as to the accuracy of such methods, total moisture in the aggregate shall be determined in accordance with ASTM C566.

When individual aggregate weigh batchers are used, the scale reading for each material shall be within 2% of the specified mass.

In a cumulative aggregate weigh batcher, the cumulative mass after each measurement shall be within 1% of the required cumulative amount when the scale is used in excess of 30% of its capacity. For cumulative measurement less than 30% of scale capacity, the allowable variation shall be ±0.3% of scale capacity or ±3.0% of the required cumulative mass, whichever is less.

Mixing water shall be measured by mass or volume to an accuracy of ±1%. Adjustments for free water contained in batched materials, such as free water in the aggregates, water contained in admixture solutions, shall be made to the quantity of water to be batched. The total amount of mixing water in the batch obtained from all sources shall be within ±3% of the specified quantity and shall not vary the designed water-to- cementitious material ratio of the mix by more than ±0.02. Mixers shall be completely emptied of all water prior to the loading of a concrete batch.

Admixtures shall be batched in liquid form by either mass or volume; in powdered form by mass or bag. Volumetric measurements of admixtures or air entraining agents shall be to an accuracy of ±3% of the required amount or 30 mL, whichever is greater. Measurement by mass shall be to an accuracy of ±3% of the required amount.

The addition of cement to a fully batched load of aggregates will not be accepted.

211.07 Mixing

211.07.01 General – Sufficient plant capacity and transporting equipment shall be provided by the Contractor to ensure continuous delivery of concrete at the rate required with the necessary intervals between batches, for the proper placing and finishing of the concrete without the formation of cold joints in the finished concrete.

211.07.02 Concrete Mixers

(a) General – Concrete mixers shall conform to the following requirements:

(i) manufacturer's rating plate carried in a prominent position, indicating rated mixing capacity, and recommended speed of mixing which must be in the range of not less than 4 nor more than 18 rpm.

(ii) demonstrated capability of operating satisfactorily at the recommended mixing speed.

(iii) number of revolutions of the drum or blades registered with a counter in working condition.

(iv) charge and discharge openings and chute free from appreciable accumulations of cement or concrete, and hopper and chute surfaces clean and smooth.

(v) batch water measuring equipment in good operating condition.

(b) Volumetric Mobile Mixers – Volumetric mobile mixers shall conform to the following requirements:

(i) shall be auger type mixers.

(ii) the mobile mixers shall be calibrated with the actual cement and aggregate products used in the concrete mix. Calibration tables and curves shall be submitted to the Ministry Representative. The volume of the concrete produced shall be confirmed and shall be within 2%.

(iii) mobile mixers shall have separate flow-controlled supply hoppers for each individual aggregate gradation and cementitious constituent material to be added volumetrically.

(iv) may be used provided that the accuracy of batching and uniformity of concrete is as required by CSA A23.1.

(v) cement flow rates shall be checked before (during if required) each placing operation.

(vi) the scale being used to calibrate the mobiles shall have a minimum capacity of 50 kg.

(vii) the water tank shall have a stand pipe showing the amount of water in the tank.

(viii) the mobiles shall have individual flow meters and admixture tanks for each admixture to be used as part of an admixture injection system.

(ix) the mobiles shall be able to advance backwards and forwards from controls located at the back of the mobile.

(x) the mixer auger shall be maintained at minimum of a 25° angle from horizontal when mixing and discharging, unless otherwise indicated by the manufacturer.

211.07.03 Mixing Concrete – Mixing drums shall be clean and empty before being charged. The drum shall be rotated at the manufacturers' recommended mixing speed during charging and mixing. Concrete shall be mixed to the uniformity requirements of CSA A23.1, Section 5.2.4.5. When a satisfactory mixing time is established, it shall be maintained for all batches of the same design and mixed with the same equipment.

211.08 Delivery

211.08.01 General – Off-site mixed concrete shall, after being mixed, be transported to the Site in either truck agitators or truck mixers operating at the speed designated by the manufacturer of the equipment as agitating speed.

When concrete is being mixed in mixer trucks while in transit, the speed of the mixing drum shall be reduced to agitating speed once the mixing time has elapsed. Delivery of mixed concrete in non-agitating equipment will not be permitted.

211.08.02 Time of Placement – Time of placement guidelines for bridge decks including diaphragms, bridge parapets, bridge medians, bridge sidewalks and approach slabs are given in SS 413, Bridge Decks and Concrete Overlays. All other concrete shall be fully discharged and placed within 120 minutes after water and cement have been combined. Under conditions contributing to rapid stiffening of concrete the Ministry Representative may require a time of less than those stated above.

Time of placement extensions will be considered on a case-by-case basis and may require the use of time extension admixtures such as Hydration Stabilizing Admixtures (HSAs) conforming to the requirements of ASTM C494 Type B, retarding or Type D, water-reducing and retarding admixtures. When accepted by the Ministry Representative, the guidelines for the use of HSAs and other admixtures to extend time of placement shall be as follows:

Modifications to the time of placement will be considered on a case by case basis and may require the use of set retarder or Hydration Stabilizing Admixtures (HSA's).

(a) Set retarder or Hydration Stabilizing Admixtures (HSAs) – Set retarders or HSAs shall only be used if approved in advance by the Ministry Representative.

When accepted by the Ministry Representative, the use of set retarders or HSA's shall be as follows:

(i) Set retarders or HSAs shall conform to the requirements of ASTM C494:

(A) Type B, Retarding Admixture; or

(B) Type D, Water-Reducing and Retarding Admixture

(ii) Concrete shall be fully discharged and placed within 180 minutes after water and cement have been combined.

(iii) When HSA's are used, these time extensions are subject to preconstruction trial testing being conducted by the Contractor to establish the appropriate HSA dosage. Preconstruction trials shall simulate expected site placement exposure conditions without increasing the water/cementitious materials ratio of the concrete above that which would be required if HSA's were not used. Sampling shall be in accordance with CSA A23.2-24C using trial batches produced by the concrete supplier of the proposed mix design.

Testing shall be carried out during preconstruction trials to verify the extended slump life and set time. Set time testing shall be carried out in accordance with ASTM C403.

(iv) The use of HSA's shall in no instance modify the maximum concrete temperature required at time of placement.

(v) A one-time only addition of HSA will be allowed; during initial batching of the concrete or immediately at completion of batching as recommended by the admixture manufacturer. Addition of HSA at any other time will be cause for rejection of the concrete.

211.08.03 Temperature Control and Thermal Control Plan – Temperature control guidelines for bridge decks including diaphragms, bridge parapets, bridge medians, bridge sidewalks and approach slabs are given in SS 413, Bridge Decks and Concrete Overlays.

The temperature of all other concrete, from the time of batching to complete discharge shall remain between 10°C and 25°C. Any concrete containing silica fume shall not have a maximum temperature in excess of 20°C at time of placement.

Mass concrete shall meet the requirements of CSA A23.1. Mass concrete shall apply to concrete placements having a minimum dimension equal to or greater that 1 m. The Contractor shall submit a thermal control plan to the Ministry Representative for review and acceptance at least 2 weeks

prior to concrete placement for mass concrete. The thermal control plan for mass concrete shall be in accordance with CSA A23.1

Methods used for maintaining temperature controls shall not diminish the quality of the concrete, and shall not alter nor exceed the specified maximum w/cm ratio.

Hot Weather Concreting shall be in accordance with SS 211.18 and Cold Weather Concreting shall be in accordance with SS 211.19.

211.08.04 Control of slump on the job site by the Addition of Water – If the measured slump is less than that specified, then up to a maximum 10% of the total designed water content, not exceeding 16 L/m^3, may be added provided all the following conditions are met:

(a) the Contractor provides documentation that confirms the mix design water-to-cementitious materials ratio would not be exceeded after addition of water on the jobsite.

(b) no more than 60 minutes have elapsed after water and cement have been combined.

(c) the concrete is in a mixer or truck mixer.

After addition of water, the drum or blades shall be turned an additional 30 revolutions or more if necessary at mixing speed. The amount of water added shall be recorded on the delivery ticket. Water shall not be added to the batch at any later time. The Contractor assumes full responsibility for the on-site addition of water and its subsequent effect on the quality of the concrete.

211.08.05 Control of Slump on the Job Site for Concrete that has High Range Water Reducer (superplasticizer) – Prior to discharge, concrete incorporating ASTM C494 Type F High Range Water Reducing or Type G High Range Water Reducing and Retarding admixtures may be retempered with water in accordance with SS 211.08.04, provided the designed w/cm is not exceeded. When concrete incorporating ASTM C494 Type F or G admixtures falls below the designated slump after discharge has begun, it shall only be retempered with Type F admixtures, not water. The amount of additional admixture added shall be recorded on the delivery ticket. The Contractor assumes full responsibility for the on-site addition of admixtures and its subsequent effect on the quality of the concrete. This addition of Type F admixture shall not be used for extending the time of placement requirements described in SS 211.08.02.

211.08.06 Control of Air Content on the Site – The air content of the concrete shall, if necessary, be brought up to the specified range by the addition of an air-entraining agent in the field. Mixing shall follow to ensure proper dispersion and upon completion of mixing, the air content shall be retested. The amount of admixture added shall be recorded on the delivery ticket. The Contractor assumes full responsibility for the on-site addition of the air entraining agent and its subsequent effect on the quality of the concrete.

211.08.07 Delivery Ticket – Concrete delivered to the Site shall be accompanied by a ticket, in either a paper or electronic format, providing the Contractor and Ministry Representative with legible information, as follows:

(a) date and serial number of ticket

(b) name and location of concrete supplier and batch plant.

(c) name of the Contractor.

(d) specific job designation (name and location).

(e) specific designation of concrete mix design.

(f) truck number, load volume, and cumulative volume of concrete in cubic metres.

(g) addition of any materials added at the Site, at the request of the Contractor including but not limited to water and all chemical admixtures.

(h) time concrete was batched, arrived at Site, and completely discharged.

(i) ordered slump or slump flow and air content.

211.09 Quality Control Inspection, Testing and Acceptance

211.09.01 General – The Contractor shall be fully responsible for hiring, scheduling, overseeing, performing and documenting all quality control testing and inspection in full compliance with the Contract.

The Ministry may perform on-site sampling and testing as a function of the Ministry's quality assurance and/or audit. Any Ministry quality testing shall not relieve the Contractor of responsibility for providing quality control.

211.09.02 Sampling and Testing – Laboratory testing shall be carried out by a testing laboratory certified in accordance with the requirements of CSA Standard A283. Field test procedures shall be undertaken by experienced personnel. All testing personnel shall be certified under CSA Standard A283 or certified as an ACI Concrete Field Testing Technician – Grade 1 (Minimum). Field inspectors shall have extensive experience in on-site quality control testing of concrete, with the applicable admixtures being used, and with on-site batch adjustments. Testing personnel shall be on-site and available to test concrete for the complete duration of any concrete placement operation. Sampling and testing shall be in accordance with the test methods and minimum frequency levels as listed in Table 211-C and Table 211-D. Any test outside the specified range shall be immediately reported to the Contractor and Ministry Representative. Copies of all test results shall be received by the Ministry Representative and the concrete supplier within one (1) day of the test date.

211.09.03 Sampling Point of Acceptance – Determination of concrete properties for acceptance will be made based on samples taken at the final discharge end of the placement

system. The Ministry Representative may accept sampling at the truck discharge end provided that testing during the test batches confirm that the concrete mix parameters are within the stated specification requirements at the actual placement point. The Ministry Representative may require additional testing to be performed at the point of placement at a predetermined frequency throughout the placement period. Such additional testing, including all costs shall be deemed to be part of the Contractor's quality control program.

211.09.04 Strength Tests – Unless otherwise specified in the Contract, a "strength test" shall be the average 28-day strength of three standard 100 mm x 200 mm test cylinders, sampled, cast, cured, transported, and tested in accordance with CSA A23.2-3C and A23.2-9C. The cylinder size shall however, meet the aggregate size limitations specified in Clause 8.2 of CSA A23.2-3C.

For each strength test, a fourth cylinder shall be cast for a 7-day strength determination, and if requested by the Ministry Representative, a fifth cylinder shall be cast for a 56-day strength determination.

If one specimen in a strength test, in the opinion of the testing agency shows evidence of improper sampling, casting, or testing, including damage from improper handling and transporting (and if agreed to by the Ministry Representative), it shall be discarded and the remaining two specimen strengths averaged.

Additional cylinders may be cast, at the discretion of the Ministry Representative or Contractor.

The strength of each classification of concrete (as defined below) shall be considered satisfactory if the following two criteria are met:

(a) the moving average of three consecutive strength tests equals or exceeds the specified strength; and

(b) each individual strength test equals or exceeds 90% of the specified strength.

Each "classification of concrete" will be taken as all of the concrete in one pay item in the Schedule 7.

Where fewer than three strength tests are taken, the results of one test or the average of two will be used.

211.09.05 Test Cylinders – Making and curing concrete test cylinders shall be carried out in accordance with CSA A23.2-3C. The test cylinders shall be cast by the Contractor using standard CSA approved moulds. The Contractor shall provide properly designed temperature-controlled storage boxes for test cylinder storage, for a period of at least 20 hours, and further protection from adverse weather and mishandling before and during delivery to the testing laboratory for curing and testing.

The Contractor shall provide a maximum-minimum thermometer for each storage box and record site curing temperatures for all test cylinders.

Storage in a portable building which will be used by the Contractor's personnel or the concrete testing personnel during the first 24 hour storage period will not be permitted. Storage facilities shall be provided, installed, and accepted before any concrete is placed.

The Contractor shall deliver the test cylinders to a CSA certified testing laboratory. Handling and transporting of the cylinders shall be in accordance with CSA A23.2-3C. No extra laboratory curing time will be allowed for cylinders that are delivered late to the laboratory. A copy of the test results shall be forwarded to the Ministry Representative within 2 days of each individual strength test date. Test results shall include all information as described in CSA A23.2, Annex B, Form for Reporting Compressive Strength of Concrete Test Cylinders.

211.09.06 Field Cured Test Cylinders – The Ministry Representative may require field cured test cylinders to be cast to check the adequacy of the Contractor's curing or cold weather protection. The Contractor may also cast field cured test cylinders for determining interim strengths that may be required for further work progress approval such as opening to traffic, formwork removal and/or erection, etc.

Casting, handling, transporting and testing shall be in accordance with CSA A23.2-3C and -9C.

Test cylinders shall be stored as near as possible to the point in the structure that the test cylinders represent, and shall be afforded the same temperature protection and moisture environment as the structure.

At the end of the curing period the test cylinders shall be left in place, exposed to the weather in the same manner as the structure.

Test results on field cured test cylinders shall not be used as a basis for acceptance or rejection of the concrete.

211.09.07 Failure to Meet Minimum Strength Specifications

(a) Understrength Concrete – When the strength of concrete is not satisfactory as specified in SS 211.09.04, the Contractor shall prepare a plan for remedial measures and the plan shall be submitted to the Ministry Representative for review and acceptance. If the proposed plan is not acceptable to the Ministry Representative, the Ministry Representative may order replacement, supplemental reinforcing, or other remedial measures, at the Contractor's expense, for portions of the structure. Alternatively, at the sole discretion of the Ministry Representative, the understrength concrete may be accepted at a reduced price.

(b) Coring – Coring of concrete to verify its strength will be allowed only for concrete which is otherwise to be replaced or reinforced. Coring will not be allowed for concrete for which reduced payment is to be made for understrength. When coring is allowed, cores shall be

taken at locations as directed by the Ministry Representative, who shall be present during the coring.

Where coring is allowed, it shall be carried out at the Contractor's expense by an independent qualified testing firm. Three cores shall be obtained and tested in accordance with CSA A23.2-14C "Obtaining and Testing Drilled Cores for Compressive Strength Testing" at each test location.

Concrete in the area represented by the core tests may be considered structurally adequate if:

(i) the average strength test result of each set of three cores from the portion of the structure in question is equal to at least 100% of the specified strength

(ii) the strength test result of any single core is not less than 80% of the specified strength

Notwithstanding the final result, the Contractor will be responsible for all costs associated with the coring operation and testing, and shall not be reimbursed for inconvenience or other associated costs.

211.09.08 Failure to Meet Slump or Air Content Specifications – In the event that the slump and/or air content tested by pressure method are outside the specified limits, the Ministry Representative may accept adjustments to correct the deficient condition as an alternative to rejection. In such cases, adjustments must be completed within the maximum time allowed as specified in SS 211.08.02 and additional testing shall be required to verify specification compliance. If compliance is not achieved, the concrete shall be rejected.

211.10 Placing of Concrete

211.10.01 General – Equipment for conveying concrete at the Site, such as buggies, buckets, hoppers, chutes, belts and pumps, shall be of such design, size and condition to deposit a continuous and adequate supply of concrete of the specified mix and consistency without segregation at the required locations.

Placing of concrete shall comply with CSA A23.1. The equipment specifications in SS 211.10.02 to SS 211.10.07 inclusive shall apply unless the use of alternative equipment is accepted by the Ministry Representative.

211.10.02 Buggies – Buggies shall have inflatable rubber tires. Inner surfaces of the bucket shall be smooth with filleted corners, mortar-tight and free of any excessive accumulation of hardened concrete, obstructions or deterioration interfering with the proper discharge of concrete. Grade and ramp-way surfaces shall be sufficiently smooth to prevent segregation of concrete being carried in buggies.

211.10.03 Bottom-dump Buckets and Hoppers – Bottom-dump buckets and hoppers shall have side slopes not less than 60° from the horizontal and shall be equipped with wide free-working and tight-closing discharge gates. The discharge gates shall be constructed to regulate the concrete flow, spring loaded to ensure complete closure and be capable of immediate closure at any time during discharge. Buckets and hoppers shall be cleaned of any accumulation of partially hardened or hardened concrete before and during concrete placement. Gate control mechanisms shall be cleaned and lubricated before being used.

In the use of crane and bucket, segregation of concrete from jarring or shaking shall be prevented. Side-dump buckets shall be used whenever necessary to avoid conflict with extended reinforcing steel or formwork.

211.10.04 Chutes – Chutes shall be of rounded cross section to avoid the accumulation of concrete in corners; be capable of slope adjustment sufficiently steep to permit flow without requiring a slump greater than that specified or required for placement (slope normally required is 2 or 2½ horizontal to 1 vertical).

Baffles and changes in direction may be used to control flow, but not vibration, paddling, or water spray.

Chutes or belts shall be rigidly supported but sufficiently mobile to permit discharge as close as possible to the placement location as it progresses. Any long line of chutes or belts shall be covered during hot, dry or windy weather to prevent drying of concrete and excessive slump loss.

211.10.05 Conveyor Operation Rate – Conveyor operation rate shall be dependent on the proper placement and consolidation of the concrete with the belt inclination such that no segregation occurs from any sliding or roll-back of the concrete.

211.10.06 Hoppers – Hoppers, set approximately level, shall be supported on specially framed bearers transmitting only vertical loads to the top of the formwork.

211.10.07 Pumps – Pumps shall be of sufficient capacity to supply un-segregated concrete, using the design mix proportions at the design slump, to meet the required placement volume at the forms. Satisfactory operation and performance of the proposed pumping equipment with the concrete materials and mixes otherwise suitable and appropriate for the job, along with the line layout and grate size opening, shall be demonstrated to the Ministry Representative's satisfaction and acceptance.

Any necessary priming of the concrete pump shall be done with a water/cement or water/cement/sand slurry. The prime slurry followed by approximately 0.25 m^3 of concrete shall not be included in the concrete placement. No other pump aid type admixtures will be permitted.

211.11 Falsework and Formwork

211.11.01 General – Formwork and falsework shall be designed, supplied, installed and removed in accordance with CSA-S269.1 and shall also meet all the requirements for concrete formwork and falsework given in the Occupational Health and Safety Regulation Clauses 20.17 to 20.26 inclusive.

Formwork and falsework support accessories shall not be welded or otherwise attached to the permanent structural steelwork or precast concrete structural members.

Where required, formwork and associated temporary falsework design and drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia and experienced in the design and construction of falsework and formwork structures similar to those required to construct the Work. No load shall be placed on the formwork or falsework until the professional engineer responsible for the design has inspected and certified, in writing, that the work has been carried out in accordance with the formwork drawings and specifications and a copy of the formwork design engineers' certification has been received by the Ministry Representative.

Bridge deck formwork design shall include a check for the lateral stability of the girders during the placement of deck concrete. Stringers shall be temporarily braced, if necessary, to facilitate deck concreting operations.

Detailed falsework and formwork drawings shall be submitted to the Ministry Representative for review. The drawings shall be submitted a minimum of two weeks before the start of installation.

211.11.02 Design – All forms shall be designed and built mortar-tight and of sufficient rigidity to prevent distortion due to the pressure of vibrated concrete and other loads incidental to the construction operation. The forms shall be substantial and unyielding, and shall be designed so that the finished concrete will conform to the design dimensions and contours. The shape, strength, rigidity, water tightness and surface smoothness of re-used forms shall be maintained at all times. Forms which are unsatisfactory in any respect shall not be used.

All formwork and falsework shall be designed to be completely removable, no stay in place formwork will be permitted.

No dry ties shall be permitted; form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling.

For narrow walls and columns, where the bottom of the form is inaccessible, removable panels shall be provided in the bottom form panel to enable cleaning out of extraneous material immediately before placing the concrete.

211.11.03 Forms for Exposed Surfaces – All forms for exposed surfaces shall be mortar-tight. All exposed edges shall be chamfered using a chamfer strip and return corners filleted. Metal bolts or anchorages within the forms shall be so constructed as to permit their removal to a depth of at least 50 mm from the concrete surface. Break-back type form ties shall have all spacing washers removed and the tie shall be broken back a distance of at least 50 mm from the concrete surface. All fittings for metal ties shall be of such design that, upon their removal, the cavities which are left will be of the smallest possible size. Torch cutting of steel hangers and ties will not be permitted. Formwork hangers for exterior surfaces of decks and curbs shall be an acceptable break-back type with surface cone, or removable threaded type. Cavities shall be filled with cement mortar and the surface left sound, smooth, even and uniform in color. See SS 211.17 for details on finishes, mortar mixes and other relevant information.

211.11.04 Deck Formwork – Formwork for decks, curbs, sidewalks and parapets shall be fabricated so that the lines and grades shown on the drawings are achieved. For suspended slab bridge decks the formwork shall be designed to allow for an adjustable haunch to facilitate the casting of a constant depth deck slab as shown on the Drawings. Girders will be erected to normally accepted standards of tolerance; it shall be necessary to adjust the formwork to compensate for camber, variances in girder dimensions, positioning, and sweep.

Prior to commencing deck formwork and subsequent to the completion of the erection of girders, with the girders in a thermally neutral condition and with no dead load other than the dead load of the girders and necessary fall protection, the Contractor shall profile all the girders. Camber elevations are to be taken at 1000 mm centres maximum (or as indicated on the Drawings or Special Provisions), centrelines of all bearings and ends of girders. The camber elevation locations are to be laid out to an accuracy of 25 mm horizontally and camber elevations taken to the nearest 1 mm vertically. Within 14 days of the Contractor supplying the camber elevation to the Ministry Representative, the Contractor will receive haunch heights for setting the deck slab soffit, and the design finished grade elevations for the deck, complete with anticipated dead load deflections at 1000 mm stations along the centreline of roadway.

If there is more than three (3) months between the camber survey and the deck pours, the camber shall be resurveyed and adjustments made to formwork and rebar as necessary to ensure an acceptable deck profile and clear cover.

In the event that actual girder camber values vary significantly from the design values, the Contractor will be required to modify the original design finished grade as directed by the Ministry Representative.

211.11.05 Removal of Falsework and Formwork – Where formwork and associated falsework have been engineered by a professional engineer, forms and their supports shall not be removed without the approval of the professional engineer responsible for the formwork and falsework design and without the acceptance of the Ministry Representative. In determining the time for the removal of falsework and formwork, consideration shall be given to the location and character of the structure, the weather and other conditions influencing the curing of the concrete, and the materials used in the mix.

Unless approved otherwise by the Ministry Representative and the Ministry's structural Professional of Record, deck overhang support brackets shall not be removed until the deck concrete has attained a minimum of concrete strength of 25 MPa.

Supports shall be removed in such a manner as to permit the concrete to uniformly and gradually take the stresses due to its own weight together with any construction loads likely to be imposed. All formwork must be removed from the completed structure.

211.12 Construction Joints

211.12.01 Construction Joints – General

(a) Definition – Joints between adjacent or successive lifts of concrete where the previously cast concrete cannot be plasticized with a vibrator, or any joint required by the Contractor's method of construction.

(b) Guidelines – Construction joints generally shall be in accordance with the following requirements:

(i) kept to a minimum.

(ii) concealed by a natural break or line in the structure.

(iii) provided for by the formwork design and placing techniques.

(c) Emergency Construction Joints – Emergency construction joints shall conform as nearly as possible to planned joints. In any case, concrete must be levelled as well as possible and, if lifts are partial, a vertical bulkhead shall be installed. Permission from the Ministry Representative shall be required to complete the cast following any emergency joint installation.

211.12.02 Joint Edges – Joint edges shall be preformed whenever possible for a true line upon form removal. Pour-strip shall be positioned relative to form ties so as to minimize form bulge at the bottom of the subsequent cast, but concrete may be cast to the top of a form instead of the pour-strip. Concrete surfaces outside of the reinforcing steel shall be steel trowelled to a true line.

211.12.03 Surfaces Preparation – Joint surfaces of the hardened concrete shall be thoroughly cleaned of foreign matter and laitance to present clean, sound surfaces with the coarse aggregate partially exposed and achieved by:

(a) brushing immediately following initial set.

(b) treatment with a surface set retardant, followed by green cutting.

(c) sand blasting.

(d) water blasting.

Where roughening of the hardened concrete is specified, the surface shall be roughened to a full amplitude of at least 5mm.

211.12.04 Bonding of Joints – Bonding of joints shall be achieved through the use of a water-cement slurry or, if specified, an epoxy-type bonding agent conforming with ASTM C 881M applied in strict accordance with the manufacturer's specifications. Slurry, when used, shall have a water-cement ratio equal to that of the concrete being placed and be brush applied to a 3 to 5 mm thickness for 100% coverage on to the cleaned existing concrete surface within five minutes of placing concrete.

211.13 Consolidation – Consolidation shall meet the requirements of CSA A23.1.

The equipment specifications in SS 211.13.01 and SS 211.13.02 shall apply unless the use of alternative equipment is accepted by the Ministry Representative.

211.13.01 Vibrators – General – The number, type, design and operation of vibrators will be subject to the acceptance of the Ministry Representative.

Vibrators shall have the following capabilities and requirements:

(a) diameter and frequency of vibrators shall conform to Table 211-H.

(b) vibration transmission to the concrete at frequencies not less than 130 Hz.

(c) vibration intensity sufficient to be visually observed on concrete with not more than 25 mm slump over a radius of at least 450 mm.

(d) sufficient number of vibrators available to properly compact each 8 m³ batch within 15 minutes after it is placed in forms, with at least two vibrators provided for each concrete placing unit.

(e) stand-by vibrator and generator available on the job at all times in case of breakdown.

Table 211-H: Requirements for Diameter and Frequency of Vibrators

Diameter (mm)	Frequency (Hz)	Rate of Placement of Concrete per Vibrator (m^3/h)	Application
20 – 40	170 – 250	1 – 4	very thin members
30 – 60	150 – 225	2 – 8	walls and slabs 200 – 300 mm
50 – 90	130 – 200	5 – 15	larger members

211.13.02 Mechanical Vibrators – Mechanical vibrators shall thoroughly consolidate concrete immediately after placing.

Vibrator usage shall be as follows:

(a) not inserted between reinforcing steel and formwork nor directly on to the reinforcing steel.

(b) inserted vertically at a rate of 75 mm per second and penetrating into the preceding layer at least 50 mm.

(c) withdrawn at a rate of 150 mm per second.

(d) grid pattern movement starting 100 mm from a formed surface and at a maximum spacing of 300 mm thereafter.

211.13.03 Form Vibrators – Form vibrators shall not be used, except at the discretion of the Ministry Representative where rebar configuration would prohibit the use of internal vibrators.

211.14 Finishing

211.14.01 Finishing Top Surfaces – Plastic concrete, struck off immediately following casting and before the accumulation of bleed water, shall be finished by hand or mechanical floating with the least possible disturbance. Water, cement or proprietary topping material shall not be added to the surface during finishing.

No steel floats or trowels shall be used on exposed surfaces.

211.14.02 Strike-off and Finishing Machines – Strike-off and finishing machines for surfacing concrete such as slabs on grade, pavements, etc. shall conform to the following:

(a) capable of accurately finishing concrete surfaces to the grade and elevation as shown on the Drawings or as directed by the Ministry Representative.

(b) surface finish production free of open texturing, plucked aggregates or projections.

(c) maintained in good mechanical repair and free of hydraulic fluid leaks, oil leaks, excessive grease and other contaminants.

211.14.03 Trowel Finishing – Trowel finishing where required:

(a) by the Drawings or specifications,

(b) for unformed surfaces exposed to view from a close distance,

(c) by close tolerance requirements (such as bearing surfaces),

shall be commenced after bleed waters have disappeared to reduce the texture of the concrete surface and produce a smooth blemish-free surface.

211.14.04 Other Textured Finishes – Other textured finishes shall be applied as and where specified.

211.14.05 Damage or Scaling – Any damage to or scaling of surfaces shall be repaired to the satisfaction of the Ministry Representative.

211.15 Curing

211.15.01 General – Freshly deposited concrete shall be protected from freezing, abnormally high temperatures or temperature differentials, premature drying, excessive moisture, moisture loss, heavy shocks, excessive vibrations and high stresses, for the period of time necessary to develop the desired properties of the concrete.

The curing period will begin following the placement of the concrete in the section cast. It shall be the responsibility of the Contractor to ensure that the system of curing and protection is properly planned, constructed and maintained throughout the entire curing period.

211.15.02 Acceptable Methods and Procedures – Acceptable curing methods and procedures shall include the following:

(a) concrete surfaces shall be cured for a period of seven consecutive uninterrupted twenty-four hour periods, however, if the Ministry Representative has accepted the use of high-early strength cement (HE), this period may be reduced as permitted by the Ministry Representative.

(b) surface temperature maintained between 10oC and 35oC, unless high-temperature curing (i.e., by hot blankets or steam) is acceptable to the Ministry Representative.

(c) hot weather and cold weather special requirements are covered by SS 211.18 and SS 211.19. The determination and recording of air and concrete temperatures to verify compliance with the requirements for hot and cold weather concreting shall be the responsibility of the Contractor. The location and methodology for the determination and recording of temperatures shall be acceptable to the Ministry Representative.

(d) formed and unformed concrete containing silica fume shall be subjected to extended wet-curing for a period of 7 days at ≥ 10 °C and for the time necessary to attain 70% of the specified strength. The curing types allowed are ponding, continuous sprinkling, absorptive mat, or fabric kept continuously wet.

(e) formed concrete protection for concrete that does not contain silica fume, with formwork left in place during the curing period (seven days), is generally not necessary except during extremes of hot or cold weather temperatures. Where formwork is removed before the end of the curing period, the concrete shall be protected in the manner specified below for unformed surfaces during the remainder of the curing period.

(f) Unformed surface protection for concrete that does not contain silica fume shall use burlap and water or where permitted, other moisture-retaining method considered

adequate to promote curing, and shall include the following common methods and requirements:

(i) Burlap – Soaking wet burlap of acceptable quality shall be carefully laid on the surface of the concrete as soon as the concrete has set sufficiently to support the burlap without marring the surface. Strips shall be overlapped 150 mm, adequately held down and maintained in place in a thoroughly wet condition throughout the curing period (seven days). Wetness of burlap shall be maintained at all times.

(ii) Vapour Barrier – Vapour barrier of acceptable quality of opaque white-pigmental 0.10 mm thick (4 mil) polyethylene sheet, white upper surfaced paper or other permitted material shall be of such size and placement to minimize the number of laps. The material shall be laid, lapped and held down around the edges and at the laps:

(A) to prevent displacement.

(B) to provide and maintain an effective vapour barrier.

(C) prevent any flow of air between the vapour barrier material and the concrete.

(D) kept in place throughout the curing period (seven days).

However, the barrier material shall not be pulled tight against the fresh concrete surface so as to smooth the required surface texture.

211.15.03 Removal of Protective Materials – Protective materials shall be removed entirely from all concrete surfaces at the end of the curing period and the surfaces left clean.

211.16 Dimensional Tolerances – Dimensional tolerances shall be as shown in Table 211-I.

211.17 Finishing Formed Surfaces

211.17.01 Class 1 Finish – Class 1 Finish is the basic finish to be produced on all formed surfaces not exposed to view unless a better finish is specified or required by the Drawings or Special Provisions.

In order to produce a Class 1 finish, the formwork shall be mortar tight. Panel marks and texture are of no importance.

All ties, bolts, nails and other metal specifically required for construction purposes shall be removed or cut back to a depth of 50 mm from the surfaces of the concrete and the resulting holes filled.

Form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling. Removable form tie rods inside of sleeves shall not be permitted.

Table 211-I: Dimensional Tolerances

ELEMENT	TOLERANCE
Trueness of bearing surfaces	such that there is no gap over 1 mm under a 1 m straight edge
Trueness of bridge deck surfaces	such that there is no gap over 3 mm under a 3 m straight edge
Trueness of slope pavement surfaces	such that there is no gap over 12 mm under a 3 m straight edge
Trueness of other surfaces exposed to view	such that there is no gap over 12 mm under a 3 m straight edge
Curbs and parapets	such that there is no gap over 3 mm under a 3 m straight edge
Horizontal distance to concrete surfaces	±50 mm
Elevations of concrete surfaces except bearings and decks	±25 mm
Elevations of bridge deck surfaces	±15 mm
Elevations of bearing surfaces	±5 mm
Variation from plumb	1:400, but not more than 30 mm
Dimensions of members over 10 m	±25 mm
Dimensions of members under 10 m	-6 mm, +15 mm
Location of anchor bolts	± 6 mm
Projections of anchor bolts, other embedments	±13 mm

***Note**: The tolerances listed in Table 211-I are acceptable only in so far as they do not prevent the proper fit of structural members.*

Honeycombs and voids over 500 mm² in area shall be filled. Honeycombs and voids shall not be repaired until inspected by the Ministry Representative as special methods of repair may be required where occurring in structural elements. Otherwise, concrete surfaces shall be repaired as follows:

(a) remove all porous concrete by cutting 20 mm into sound concrete.

(b) thoroughly clean reinforcing steel without damaging same.

(c) cut edges of cavities as nearly perpendicular to the concrete surface as possible and deep enough to hold the patching mortar.

(d) keep surrounding concrete saturated for 30minutes before patching.

(e) after the sheet of free water has disappeared from the concrete surfaces, apply a latex bonding agent, in strict accordance with the manufacturer's instructions.

(f) patch with mortar, proportioned as specified in SS 211.17.05 and mixed with the minimum amount of water necessary to make a workable paste: for voids deeper than 10 mm, multiple layers of mortar each with a maximum thickness of 10 mm shall be used with 30 minute intervals allowed between layers.

(g) trowel the surface of the patch flush with the surface of the surrounding concrete.

211.17.02 Class 2 Finish – Class 2 Finish is to be produced on all formed surfaces exposed to view from a moderate distance, such as surfaces of abutments and piers, and to any surfaces for which a Class 2 finish is specified or required by the Drawings or the Special Provisions. A Class 2 finish shall provide surfaces of uniform colour and texture as viewed from 25 m.

In order to produce a Class 2 finish, formwork shall be mortar tight and shall render a true surface. Fins 3 mm wide (maximum) shall be allowed at the panel joints; however, sheathing joints must be mortar tight. Irregularities of 3 mm in height with areas of 50 mm x 75 mm shall be allowed to a maximum of four such areas per 3 m² of formwork. Patches of dissimilar material will not be permitted. Horizontal and vertical joints shall be aligned.

All ties, bolts, nails and other metal specifically required for construction purposes shall be removed or cut back to a depth of 50 mm from the surfaces of the concrete and the resulting holes filled.

Form tie rods shall remain embedded in the concrete and terminate not less than 50 mm from the formed face. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling. Removable form tie rods inside of sleeves shall not be permitted.

Honeycombs and voids over 25 mm diameter shall be filled, and all bug holes over 5 mm diameter shall be pointed. All fins and projections shall be removed with a hand stone or power grinder. The use of a power grinder shall be kept to a minimum and confined to the areas required.

When a rubbed finish is not called for, patches shall be textured with a mortar float or lightly brushed after trowelling smooth.

Where more than 50 voids or bug holes over 5 mm diameter occur per square metre, or if the surfaces are not acceptably uniform in colour or texture, the entire area affected shall be given a rubbed finish, as follows:

(a) allow patches to attain an initial set.

(b) keep surfaces saturated with water for 60 minutes before applying mortar.

(c) remove free water from surfaces.

(d) apply mortar, proportioned as specified in SS 211.17.05 and mixed at least 60 minutes before application, to the concrete surfaces with a sponge, float or a hand carborundum stone, and work mortar well into the surfaces.

(e) allow mortar to dry 30 to 60 minutes.

(f) remove residues from the surfaces by rubbing with clean dry burlap but with no addition of neat cement to the surface during the "sacking" operation and finally cure the surfaces as specified in SS 211.15.

211.17.03 Class 3 Finish – Class 3 Finish is to be produced on all formed surfaces exposed to view from close distances, such as surfaces of curbs and parapets, and to any surface for which a Class 3 finish is specified. A Class 3 finish shall provide surfaces of uniform colour and texture when viewed from 10 m.

In order to produce a Class 3 finish, the formwork shall render a true smooth surface, free from fins and projections. New plywood or steel is necessary to produce the required finish. Re-use of plywood forms will be permitted only if in an "as new" condition. Repairs to the forms shall be with full panels of sheathing only.

All ties, bolts, nails and other metal specifically required for construction purposes shall be removed or cut back to a depth of 50 mm from the surfaces of the concrete and the resulting holes filled.

Form tie rods shall remain embedded and terminate not less than 50 mm from the formed face of the concrete. Removable embedded fasteners on the ends of the rods shall be such as to leave holes of a regular shape for reaming and filling. Removable form tie rods inside of sleeves shall not be permitted.

Patching shall be:

(a) patch installation projecting slightly from the surrounding concrete surface.

(b) after 24 hours, saturated and carefully finished to render the surfaces of the patch true to the surrounding concrete, but any grinding using a hand carborundum stone kept to a minimum.

When a rubbed finish is required, the applied mortar shall be carefully worked into the surface area being finished and, using a hand carborundum stone manipulated in a circular motion, the entire surface rendered true and smooth with all excess mortar removed.

211.17.04 Class 3 Finish Alternative Procedures – Class 3 Finish alternative procedures when, in the opinion of the Ministry Representative, the formwork may safely be removed at an early age and the concrete finished while still partly plastic. Procedures shall include the following:

(a) careful removal of the formwork when the concrete has sufficiently hardened so as to hold its shape.

(b) finishing the surfaces immediately (within 4 to 8 hours of casting, depending upon weather).

(c) careful rubbing of the concrete surfaces with a carborundum stone, occasionally wetting same to produce a paste from the surface of the concrete.

(d) rubbing continued, adding necessary sand-cement mortar in small quantities to fill voids, until all voids and form marks are removed.

(e) surface curing as specified.

Modifications to the surface texture shall be achieved by grinding or other means.

211.17.05 Patching Mortar – Patching mortar where specified above shall be selected from the Ministry Recognized Products List, subject to the acceptance of the Ministry Representative.

211.18 Hot Weather Concreting – When ambient air temperature is 25°C or higher or there is a probability of it rising to 25°C during the placing period (as forecast by the nearest official meteorological office), hot weather concreting procedures as outlined below, shall be used, for all concrete other than deck concrete. Additional requirements for hot weather concreting for bridge decks including diaphragms, bridge parapets, bridge medians, bridge sidewalks and approach slabs are given in SS 413, Bridge Decks and Concrete Overlays.

Prior to the placement of concrete, the Contractor shall submit to the Ministry Representative for acceptance, the proposed placing operations complying with the requirements for hot weather concreting.

Curing shall be accomplished by water spray or by using saturated absorptive material, such as burlap.

Formwork, reinforcement and concreting equipment shall be protected from the direct rays of the sun or cooled by fogging and evaporation.

Concrete for piers, abutments and footings of bridges shall not have a temperature greater than 25°C at any time prior to placement and curing.

The initial temperature of concrete prior to placement may be estimated from the temperatures of its ingredients by using the following equation:

$$T = \frac{0.22(T_a M_a + T_c M_c) + T_w M_w + T_{wa} M_{wa}}{0.22(M_a + M_c) + M_w + M_{wa}}$$

where

T_a, T_c, T_w and T_{wa}

= temperature in °C of aggregates, cementitious materials, added mixing water, and free water on aggregates, respectively

M_a, M_c, M_w and M_{wa}

= mass in kilograms of aggregates, cementitious materials, added mixing water, and free water on aggregates, respectively

If deemed necessary, the Contractor shall employ the following hot weather concreting procedures to reduce the concrete temperature:

(a) Mixing Water

(i) Shading of storage tanks and water supply lines;

(ii) Refrigeration of mixing water; and/or

(iii) Adding cubed, shaved or chipped ice directly to the concrete mixer drum, and ensuring ice is of a particle size to be completely melted before the concrete is discharged from the mixer drum for placement.

(b) Cement

(i) Maintain all cementitious material to below a maximum temperature of 50°C

(c) Aggregate

(i) Shading of coarse and fine aggregate stockpiles; and/or

(ii) Wetting of coarse aggregate and allowing time for drainage of free water.

(iii) Wetting down of stockpiles of fine aggregate is not permitted.

(d) Mixing – Mixing time should be kept to a minimum, allowing adequate time for complete mixing of the concrete. The external surface of the mixing drum should be frequently wetted down.

(e) Placing – Production of concrete during hot weather shall be scheduled for a continuous supply of concrete at the Site.

Adequate manpower, equipment and standby provisions shall be provided in order to place and consolidate the delivered concrete within 45 minutes after initial mixing. Further, the time interval between placing batches shall not exceed 30 minutes.

Time of placement extensions during hot weather concreting may be considered by the Ministry Representative and may require the use of Hydration Stabilizing Admixtures (HSA's) conforming to the requirements of ASTM C494 Type B, Retarding or

Type D, Water-Reducing and Retarding Admixtures. Use of HSA's shall be in accordance with SS 211.08.02.

For large unformed surfaces, the quantity and rate of placing of concrete shall be dependent upon the rate of finishing and climatic condition.

When the rate of evaporation from unformed surfaces as estimated from SS Drawing SP211-01 "Surface Evaporation Rate" exceeds 1.0 kg/m²/hr, either concrete operations shall cease or, upon acceptance of the Ministry Representative, necessary precautions taken to prevent plastic shrinkage as detailed in CSA A23.1 Section 7.6.

(f) Curing – Curing during hot weather shall be commenced as soon as possible.

Forms shall be kept moist and loosened as soon as this can be done without structural damage, with a flow of water introduced and allowed to run down inside.

Newly exposed concrete surfaces shall be protected from premature drying by the application of pre-wetted curing materials or a continuous water spray as the forms are removed.

Large exposures of concrete require special precautions as follows:

(i) burlap application commenced as soon as the surface will support the mass without excessive deformation, with this surface kept continuously wet by fog spraying until a waterproof barrier is securely in place.

(ii) concrete under the curing blanket shall not be allowed to dry out until a minimum of seven days have passed from the time of initial covering, necessitating frequent checking and soaking by the addition of water under the top evaporation barrier.

(iii) wet curing followed by the removal of the evaporation barrier only under such conditions that the wetted burlap is permitted to dry slowly before removal and the surface is exposed to prevent excessive thermal shock.

211.19 Cold Weather Concreting – Cold weather concreting procedures shall be used when the ambient temperature is, or is forecast to be, below 5°C during placement, and/or is forecast to fall below 5°C during the first seven days after placing. The Contractor shall be fully responsible for the protection of concrete during cold and adverse weather conditions and shall maintain a minimum concrete temperature of 10°C for a minimum of seven continuous days. Prior to the placement of concrete, the Contractor shall submit to the Ministry Representative for acceptance, the proposed placing operations complying with the requirements for cold weather concreting.

211.19.01 Provisions – The following provisions for cold weather concreting shall apply:

(a) Material Temperatures: The temperature of the combined water and aggregates shall not exceed 40°C when combined with the cement. The temperature of the concrete shall be between 10°C and 25°C at the time of placement.

(b) Aggregates: Aggregates shall be heated to eliminate frozen lumps, ice and snow without overheating or excessive drying. Aggregates shall not be heated above 65°C, and all lumps of frozen aggregate shall be excluded from the mix.

(c) Formwork, Reinforcing Steel: Before any concrete is placed, all ice, snow or frost shall be completely removed from the forms and the temperature of contact surfaces raised to a minimum of 5°C, with such minimum established and maintained for at least one hour prior to placement.

Heat shall be applied uniformly and at a rate which will not induce excessive thermal stresses in the section being heated. Ambient air temperatures shall not exceed concrete temperatures by more than 13°C and shall be raised at a maximum rate of 2°C/hr.

Projecting reinforcing steel shall be insulated when temperatures are between 0°C and -10°C for a minimum of 300 mm away from the fresh concrete and for a minimum of 600 mm when temperatures are below -10°C.

(d) Other Contact Surfaces: Concrete shall not be placed on frozen surfaces except in pile founded footings where a minimum layer of 25 mm rigid insulation shall be installed. Rock surfaces shall be heated by forced air. Heating shall continue until all standing frost crystals are dissolved. Frozen embankment shall be thawed and re-compacted prior to placing concrete thereon. The temperature of existing concrete work shall be raised to a minimum of 5°C and maintained at this temperature for a time sufficient to raise the entire mass of concrete to this temperature.

(e) Curing and Protection: Curing and protection necessary to maintain a minimum concrete temperature (10°C) shall be provided throughout the curing period.

Heated enclosures shall be kept at 95% minimum relative humidity. Concrete surfaces shall be protected by formwork or impermeable membranes from direct exposure to the combustion gases of heating. The use of salamanders, coke stoves, oil or gas burners and similar spot heaters which have an open flame and intense local heat is prohibited without the Ministry Representative's acceptance. The enclosed air temperature shall not be less than 10°C nor more than 26°C.

Where practical, insulated forms, capable of maintaining the surface of the concrete at not less than 10°C for a period of seven days, may be used instead of enclosures and heating. If forms are insulated, exposed horizontal surfaces shall be protected with a similar layer of the insulating material securely fastened in place. If the insulated forms do not maintain the proper temperature at the concrete surface, auxiliary protection shall be used to provide additional heat as described above.

The determination and recording of air and concrete temperatures to verify compliance with the requirements for cold weather concreting shall be the responsibility of the Contractor.

The Contractor is responsible for furnishing and using a 24-hour minimum/maximum or continuous temperature recording thermometer to record the air temperature within the enclosure and/or insulated forms. Daily records shall be submitted to the Ministry Representative upon request.

At the end of the curing period, heating and protection shall be withdrawn in such a manner so as not to induce thermal shock stresses in the concrete. The temperature of the concrete shall be gradually reduced at a rate not exceeding 10°C per day to that of the surrounding air. To achieve this, in a heated housing, the heat shall be slowly reduced and then shut off, and the whole housing allowed to cool to air temperature before the housing itself is removed. However, the protection shall not be removed until the temperature of the concrete has fallen to within 10°C of the temperature of the outside air.

211.20 Underwater and Tremie Concreting – Underwater concreting procedures shall be used when structures, or parts of structures, are to be constructed under water. This section is also applicable to tremie placement in the dry when specified.

211.20.01 Requirements – The concrete, methods of placement either by pumping or tremie pipes, and equipment shall be in accordance with the following, or as otherwise directed by the Ministry Representative:

(a) Proposals outlining procedures shall be submitted for review to the Ministry Representative 3 weeks prior to placement of concrete. Placement shall comply with the provisions of the proposal.

(b) Materials shall conform to the requirements of SS 211.04, except that coarse aggregates shall contain not more than 25% angular particles by mass. The use of 14 mm Nominal Maximum Size of Coarse Aggregate is permissible for tremie concrete.

(c) Mix proportions shall be as specified in SS 211.03, except:

(i) water/cement ratio no greater than 0.40.

(ii) maximum plasticized slump shall be 190 mm ±40 mm. The slump test as per A23.2-5C shall be supplemented by the slump flow test of A23.2-19C. The concrete sample for A23.2-19C shall be consolidated as described in A23.2-5C prior to measuring the slump flow. The Visual Stability Index (VSI) shall be 0 or 1 determined in accordance with CSA A23.2-19C.

(iii) admixtures used to achieve plastic, cohesive and flowable concrete if acceptable to the Ministry Representative.

(iv) admixtures shall be used as required to eliminate excessive bleeding and segregation.

(v) retarding type admixture if accepted by the Ministry Representative shall be used to prevent the formation of cold joints with the lift method of placement or where the concrete surface area is large.

(d) Heating of water or aggregates or both, to obtain a suitable placing temperature, shall be done in accordance with SS 211.19. Concrete, when mixed, shall have a temperature of between 15°C and 25°C and shall not be placed in water having a temperature less than 2°C.

(e) Cofferdams or forms into which concrete is to be placed shall be sufficiently tight to prevent the loss of mortar and, if in running water, reduce the velocity of flow within the cofferdam or forms to not more than 3 m per minute. Dewatering will not be permitted until at least 24 hours after concreting has been completed.

(f) Pumping and tremie pipe requirements shall be as follows:

(i) length sufficient to reach the lowest point of deposit but held 150 mm to 250 mm above the bottom of the placement.

(ii) submerged into the placed concrete and held in a near vertical position at all times.

(iii) supported in such a manner to allow vertical movement.

(iv) discharge at locations not farther apart than 7 m.

(v) discharge end fitted with a 3 m section of steel pipe.

(vi) fed by either chutes or pumps with a continuous flow of concrete.

(g) Priming of tremie pipes shall be by inserting a plug of burlap, 25 mm less in diameter than the tremie pipe, in the top of the pipe and carefully controlling concrete flow as the plug passes through the pipe. Where priming is lost, the priming procedure shall be repeated.

(h) Concrete flow, once begun, shall be continuous through the pipe, at a minimum of 15 m^3 of concrete supplied to each tremie or pump per hour, but controlled so as not to overflow the feed hopper. The flow of concrete shall be continuous to the end of the batch and the pipe lowered in the placed concrete to maintain its prime.

(i) Placement of concrete underwater shall be either to the full depth of placement or in lifts and shall conform to the following requirements:

(i) seals shall be placed by beginning at one end and continuing until final grade is achieved.

(ii) point of placement shall be relocated by approximately 3 m, either laterally or longitudinally and placement continued to grade.

(iii) above procedures repeated to the completion of the seal or when the lift method of placement is used.

(iv) flow of concrete shall be regulated to produce approximately horizontal surfaces.

(v) each lift placed before the preceding lift has taken initial set.

(vi) depth and area of each lift dependent upon the rate of placement.

(j) Concrete in its plastic state shall not be disturbed either directly or indirectly by puddling or vibration.

(k) Should interrupted placement be sufficient for the concrete to take initial set, the laitance shall be removed from the concrete surface before concreting is resumed, but with the resumption of concreting delayed indefinitely, laitance shall be removed not later than 36 hours after the interruption.

211.21 Measurement and Payment

211.21.01 Formwork – Payment for formwork will be made at the Unit Prices bid per square metre. Payment shall be for the supply, construction and removal of formwork, and for the finishing, as required, of formed surfaces. Areas shall be measured of the faces of forms in contact with the concrete.

Payment will be made for forming one face of vertical construction joints which are shown on the Drawings or ordered by the Ministry Representative. No payment will be made for forming construction joints not so required. Payment will be made for forming both faces of joints formed by casting in sheets of styrofoam or other material, where such joints are required by the Contract Documents.

Payment will not be made for forming surfaces which are cast against permanent embedded items which are paid for separately.

Upper surfaces which form an angle of 45° or less with the horizontal, will not usually be considered to be formed surfaces.

The undersides of members which are shown on the Drawings to rest on ground or fill will not be considered formed surfaces. The undersides of members which are not shown on the Drawings to rest on ground or fill will be considered formed surfaces.

Where sides of footings or faces of walls are cast against ground or fill (if permitted by the Ministry Representative) the surfaces so formed will be considered formed surfaces. Only the areas of the surfaces shown on the Drawings will be paid for.

Progress payments will be made according to Table 211-J.

Table 211-J: Progress Payments

Progress Payments	Application
60%	For formwork in place
20%	For Formwork removed
20%	For surface finishing completed

211.21.02 Concrete – Payment for concrete will be made at the Unit Prices bid per cubic metre (or at a reduced price for understrength concrete - see fourth paragraph of this subsection) prorated at 90% for the supply, transportation, placing, and finishing (except the finishing of formed surfaces) of concrete, and for everything supplied and done in connection therewith, except formwork and reinforcing steel.

Payment for curing, prorated at 10% will be made pending satisfactory completion of the specified curing process. No payment will be made for partial curing. Failure to comply strictly with the specified curing process will result in forfeiture of payment for curing. The Ministry Representative shall take measures as required to provide the specified curing. Should the cost of these measures exceed the monies recovered from the forfeiture of curing payment, the balance will be deducted from the remaining payment for concrete.

Payment for concrete shall include cement.

Payment shall also include all costs associated with quality control.

Measurement of volume will be made to the surfaces shown on the drawings or ordered by the Ministry Representative. Deductions will be made of the volumes of embedded ducts, timbers, etc., but not of the volume of reinforcing steel. No payment will be made for concrete placed outside the surfaces shown or ordered, as to fill spaces left by over excavation or irregularities in excavation.

The reduced price to be paid for understrength concrete shall be calculated by the following formula:

$$P_R = \frac{S_{AV}}{S_{SP}} \times P_U$$

Where

P_R = reduced price

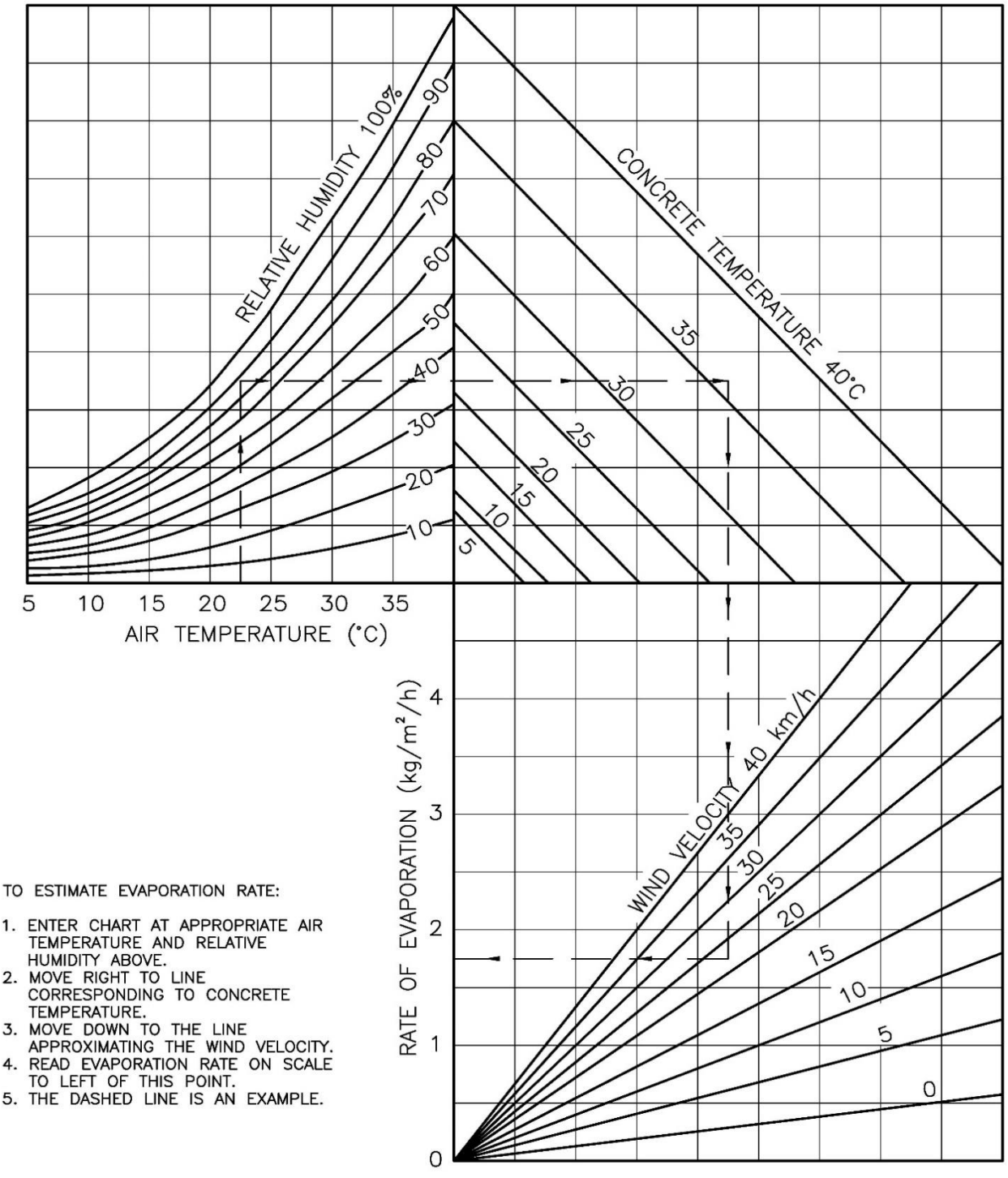
P_U = bid price

S_{SP} = specified strength

S_{AV} = the lowest average of three consecutive strength tests for one classification of concrete within the same structural element. Where less than three strength tests are taken, the results shall be the average of all the actual strength tests taken.

See SS 211.09.07 for circumstances under which a reduced price will be paid. Where a reduced price is paid, it will be applied to all of the concrete in the applicable pay item.

SURFACE EVAPORATION RATE	SP211–01



TO ESTIMATE EVAPORATION RATE:

1. ENTER CHART AT APPROPRIATE AIR TEMPERATURE AND RELATIVE HUMIDITY ABOVE.
2. MOVE RIGHT TO LINE CORRESPONDING TO CONCRETE TEMPERATURE.
3. MOVE DOWN TO THE LINE APPROXIMATING THE WIND VELOCITY.
4. READ EVAPORATION RATE ON SCALE TO LEFT OF THIS POINT.
5. THE DASHED LINE IS AN EXAMPLE.